Solution 1

We know that the optimal Θ through the normal equations is given by

$$\Theta = (X^T X)^{-1} X^T Y$$

Since $y_{ij} \in \mathbb{R}$ & $x_i \in \mathbb{R}$, $i = 1, \ldots, 720$; $j = 1, 2$.
θ'_F & θ'_J should $\in \mathbb{R}$ & therefore θ_F^T & $\theta_J^T \in \mathbb{R}$.
$\therefore \theta_F = \theta_F^T$ & $\theta_J = \theta_J^T$.

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{720} \end{bmatrix}_{720\times 1} \qquad X^T = \begin{bmatrix} x_1 \ldots x_{720} \end{bmatrix}_{1\times 720}$$

$$Y_1 = \begin{bmatrix} y_{11} \\ y_{21} \\ \vdots \\ y_{720,1} \end{bmatrix}_{720\times 1} \qquad Y_2 = \begin{bmatrix} y_{12} \\ y_{22} \\ \vdots \\ y_{720,2} \end{bmatrix}_{720\times 1}$$

$y_i = (y_{i1}, y_{i2})$

so $Y = \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \\ \vdots & \vdots \\ y_{720,1} & y_{720,2} \end{bmatrix}_{720\times 2}$

$x_i \in \mathbb{R}$, so Θ^T is a 2×1 matrix.
So, θ' & y_i are 2×1 matrices & Θ is a 1×2 matrix.
In $(X^T X)^{-1} X^T Y_i$:
$X^T X$ is a 1×1 matrix so $(X^T X)^{-1} \in \mathbb{R}$.
Also $X^T Y_i \in \mathbb{R}$.
Now,

$$\theta_F = (X^T X)^{-1} X^T Y_1$$
$$\theta_J = (X^T X)^{-1} X^T Y_2$$

$\Theta = (X^T X)^{-1} X^T Y$
Here $X^T Y$ is a 1×2 matrix,

$$X^T Y = [x_1 \ldots x_{720}] \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \\ \vdots & \vdots \\ y_{720,1} & y_{720,2} \end{bmatrix} = [X^T Y_1 \quad X^T Y_2]$$

Thus $\Theta = [(X^T X)^{-1} X^T Y_1 \quad (X^T X)^{-1} X^T Y_2] = [\theta_F \quad \theta_J]$
$\therefore \Theta = \Theta_{FJ}$

Gradient Descent Equivalence

If everyone uses gradient descent,
Cost function $J(\Theta) = \sum_i \left\| \begin{bmatrix} y_{i1} \\ y_{i2} \end{bmatrix} - (\Theta^T x_i + \theta') \right\|^2$

$$\begin{aligned} &= \sum_i \begin{bmatrix} y_{i1} - \theta_F x_i - \theta'_F \\ y_{i2} - \theta_J x_i - \theta'_J \end{bmatrix}^T \begin{bmatrix} y_{i1} - \theta_F x_i - \theta'_F \\ y_{i2} - \theta_J x_i - \theta'_J \end{bmatrix} \\ &= \sum_i (y_{i1} - (\theta_F x_i + \theta'_F))^2 + (y_{i2} - (\theta_J x_i + \theta'_J))^2 \\ &= J(\theta_F) + J(\theta_J) \end{aligned}$$

Gradient descent:

$$\theta_F^{(t+1)} = \theta_F^{(t)} - \alpha \nabla_{\theta_F} J(\theta_F^{(t)})$$
$$\theta_J^{(t+1)} = \theta_J^{(t)} - \alpha \nabla_{\theta_J} J(\theta_J^{(t)})$$

$$\Theta^{(t+1)} = \Theta^{(t)} - \alpha \begin{bmatrix} \nabla_{\theta_F}(J(\theta_F)^{(t)} + J(\theta_J)^{(t)}) = \nabla_{\theta_F} J(\theta_F^{(t)}) \\ \nabla_{\theta_J}(J(\theta_F)^{(t)} + J(\theta_J)^{(t)}) = \nabla_{\theta_J} J(\theta_J^{(t)}) \end{bmatrix}$$

$\therefore$ The gradient descent algorithm can produce the same result as the Jewellers & flower seller's algorithm provided we start with the same initial parameter values, and use the same learning rate & loss function.

Solution 2

(a)

From Mercer's Theorem:
we must show $K(x, y)$ is too a positive semi definite & symmetric.

$$K(x, y) = (x^T py + c)^2 = (y^T x + c)^2 = K(y, x)$$

$\therefore K(x, y)$ is symmetric

$$x^T y = \sum_{i=1}^{n} x_i y_i$$

$$\Rightarrow (x^T y + c)^2 = \left(\sum_{i=1}^{n} x_i y_i + c \right)^2$$

Using Multinomial Theorem,

$$(a_1 + a_2 + \ldots + a_m)^\ell = \sum_{k_1 + \ldots + k_m = \ell} \frac{\ell!}{k_1! \cdots k_m!} \prod_{t=1}^{m} a_t^{k_t}$$

So, $(x^T y + c)^2 = \displaystyle\sum_{k_1 + \ldots + k_{n+1} = \ell} \frac{\ell!}{k_1! \cdots k_{n+1}!} (x_1 y_1)^{k_1} \cdots (x_n y_n)^{k_n} \, c^{k_{n+1}}$

$$= \sum_{k_1 + \ldots + k_{n+1} = \ell} \frac{\ell! \, c^{k_{n+1}}}{k_1! \cdots k_{n+1}!} \left(x_1^{k_1} x_2^{k_2} \cdots x_n^{k_n} \right) \left(y_1^{k_1} \cdots y_n^{k_n} \right)$$

$$\frac{\ell! \, c^{k_{n+1}}}{k_1! \cdots k_{n+1}!} = \text{Ⓐ} \quad \text{(const.)}$$

$$c \geq 0 \quad \text{so} \quad A \geq 0.$$

for a particular $k_1, \ldots, k_{n+1}$,

$$(x_1^{k_1} \cdots x_n^{k_n}) = \phi_k(x)$$

$$(y_1^{k_1} \cdots y_n^{k_n}) = \phi_k(oy) \quad \text{[corresponding to } y\text{]}$$

So $k(x, y) = \displaystyle\sum_{k_1 + \ldots + k_{n+1} = \ell} A \, \phi_k(x) \cdot \phi_k(y)$

$$= \sum_{k_1 + \ldots + k_{n+1} = \ell} \sqrt{A} \, \phi_k(x) \cdot \sqrt{A} \, \phi_k(y)$$

$\Rightarrow \sqrt{A} \, \phi_k(x)$ contains all possible monomials of the form $\sqrt{A} \;\; x_1^{k_1} \times \cdots \times x_n^{k_n}$.

$$\therefore \quad k(x, y) = (x^T y + c)^2 = \sum_{k_1 + \ldots + k_{n+1} = d} \phi'_k(x) \, \phi'_k(y)$$

$$= \langle \phi'(x), \, \phi'(y) \rangle$$

where $\phi'(x) = \sqrt{A} \, \phi_k(x)$
and $\phi'_k(x)$ is the kth element of $\phi'(x)$.
$\phi(x)$ is as many elements as the # possible values of k.

$$\sum_i \sum_j z_i \bar{z}_j \sum_k \phi'_k(x_i) \, \phi'_k(y_j)$$

Let x be finite set of points $\{x_1, \ldots, x_n\}$

$$z^T K(x_{ij}) z = \sum_{i,j} z_i \bar{z}_j \, k(x_i, x_j)$$

$$= \sum_{i,j} z_i \bar{z}_j \sum_k \phi'_k(x_i) \, \phi'_k(x_j)$$

$$= \sum_k \left(\sum_i z_i \phi'_k(x_i) \right) \left(\sum_j z_j \phi'_k(x_j) \right)$$

$$= \sum_k \left(\sum_i z_i \phi'_k(x_i) \right)^2$$

$$\geq 0 \qquad (\because \text{ sum of squares})$$

$\therefore$ $K(x, y)$ kernel matrix is +ve semi-definite.
$\therefore$ $k(x, y)$ is a valid kernel.

(b)
By Mercer's Theorem, for k to be a valid kernel, the kernel matrix

$$K(x, x) \geq 0$$

$$= (x^T x)^2 - x^T x \geq 0$$

$$= (x^T x)(x^T x - 1) \geq 0$$

So if $x^T x \neq 0$ and $x^T x < 1$ then this will not hold true.
Taking $z = 1$ and $x_1 = 0.1$

$$z^T k(x, x) z = 1 \times 1 \times k(0.1, 0.1)$$

$$= (0.1 \times 0.1)^2 - (0.1 \times 0.1)$$

$$= 0.0001 - 0.0101$$

$$= -0.0099$$

$$\leq 0$$

$\therefore$ $k(0.1, 0.1) \leq 0$
$\therefore$ K is not +ve semi-definite
$\therefore$ K is not a valid kernel.

(c)
Since $K_1(x, y)$ & $K_2(x, y)$ are valid kernels, by Mercer's Theorem,

$$\sum_{i=1}^{n} \sum_{j=1}^{n} z_i z_j \, k_1(x_i, x_j) \geq 0$$

$$\sum_{i=1}^{n} \sum_{j=1}^{n} z_i z_j \, k_2(x_i, x_j) \geq 0$$

For $k = k_1 + k_2$

$$\sum_{i=1}^{n} \sum_{j=1}^{n} z_i z_j \, k(x_i, x_j)$$

$$= \sum_{i=1}^{n}\sum_{j=1}^{n} z_i z_j \left(k_1(x_i, x_j) + k_2(x_i, x_j)\right)$$

$$= \sum_{i=1}^{n}\sum_{j=1}^{n} z_i z_j \, k_1(x_i, x_j) + \sum_{i=1}^{n}\sum_{j=1}^{n} z_i z_j \, k_2(x_i, x_j)$$

$$\geq 0 \qquad (\because \text{ sum of two non-negative numbers})$$

Also $k(x, y) = k_1(x, y) + k_2(x, y)$

$$= k_1(y, x) + k_2(y, x) \qquad [\because K_1 \& K_2 \text{ are valid kernels}]$$

$$= k(y, x)$$

$\therefore K$ is symmetric
$\therefore K$ is a valid kernel

If $k(x, y) = c\, k_1(x, y)$, $c \geq 0$
$\exists$ feature maps ϕ_1 & ϕ_2 corresponding to K_1 & K_2, s.t.

$$k_1(x, y) = \langle \phi_1(x),\, \phi_1(y) \rangle = \phi_1(x)^T \phi_1(y)$$

$$k_2(x, y) = \langle \phi_2(x),\, \phi_2(y) \rangle = \phi_2(x)^T \phi_2(y)$$

Now $\phi_1(x)^T \phi_1(y) + \phi_2(x)^T \phi_2(y)$

$$= \begin{bmatrix} \phi_1(x)^T & \phi_2(x)^T \end{bmatrix} \begin{bmatrix} \phi_1(y) \\ \phi_2(y) \end{bmatrix}$$

$$= \begin{bmatrix} \phi_1(x) \\ \phi_2(x) \end{bmatrix}^T \begin{bmatrix} \phi_1(y) \\ \phi_2(y) \end{bmatrix}$$

$\therefore$ Define $\phi(x) = \begin{bmatrix} \phi_1(x) \\ \phi_2(x) \end{bmatrix}$

Feature map of $k(x, y) = \phi(x) = \begin{bmatrix} \phi_1(x) \\ \phi_2(x) \end{bmatrix}$

If $k(x, y) = c\, k_1(x, y)$,

$$\sum_{i=1}^{n}\sum_{j=1}^{n} z_i z_j \underbrace{k(x_i, x_j)} = \sum_{i=1}^{n}\sum_{j=1}^{n} z_i z_j \, c \underbrace{k_1(x_i, x_j)}$$

$$= c \left(\sum_{i=1}^{n}\sum_{j=1}^{n} z_i z_j \, k_1(x_i, x_j) \right) \geq 0$$

[$\because$ product of a +ve and a non-negative no.]

$$k_1(x, y) = \phi_1^T(x)\, \phi_1(y)$$

$$c\, k_1(x, y) = c\, \phi_1^T(x)\, \phi_1(y) = \sqrt{c}\, \phi_1^T(x) \cdot \sqrt{c}\, \phi_1(y)$$

$\therefore$ Redefine $\phi(x) = \sqrt{c}\, \phi_1(x)$
Then $\phi(x)^T \phi(y) = (\sqrt{c}\, \phi_1(x))^T (\sqrt{c}\, \phi_1(y))$

$$= c\, \phi_1(x)^T\, \phi_1(y)$$

Feature map of $k(x, y) = c\, k_1(x, y)$ is $\phi(x) = \sqrt{c}\, \phi_1(x)$

(ii) $K_1(x, y)$ & $K_2(x, y)$ are valid kernels.

So, the kernel matrices are +ve semi-definite. $(K_1)_{ij}$ & $(K_2)_{ij}$
$k(x, y) = k_1(x, y)\, k_2(x, y)$ so the kernel matrix for K is

$$K_{ij} = k(x_i, x_j) = k_1(x_i, x_j)\, k_2(x_i, x_j)$$

$$= (K_1)_{ij}\, (K_2)_{ij}$$

$$= (K_1 \circ K_2)_{ij}$$

$\therefore K = K_1 \circ K_2$

By Schur product theorem, the entrywise product (A.M) of two positive semi-definite matrices is also positive semi-definite.

$\therefore K$ is +ve semi-definite

Also $k(x, y) = k_1(x, y)\, k_2(x, y)$

$$= k_1(y, x)\, k_2(y, x) \qquad [\because K_1 \& K_2 \text{ are valid kernels}]$$

$$= k(y, x)$$

$\therefore K$ is symmetric

$\therefore$ By Mercer's Theorem, K is a valid kernel.

(iii) $k_n(x, y) = k_n(y, x)$ $[\because K_n(x, y)$ is a valid kernel$]$

$$\lim_{n\to\infty} k_n(x, y) = \lim_{n\to\infty} k_n(y, x)$$

$$\Rightarrow k(x, y) = k(y, x)$$

$\therefore K$ is symmetric

We know that $\forall\, n$,

$$\sum_{i=1}^{n}\sum_{j=1}^{n} z_i z_j\, k_n(x_i, x_j) \geq 0 \qquad [\text{+ve semi definite}]$$

$$\Rightarrow \lim_{n\to\infty}\left(\sum_{i=1}^{n}\sum_{j=1}^{n} z_i z_j\, k_n(x_i, x_j)\right) \geq 0$$

This is a finite sum ($n \times n$ terms), so

$$\sum_{i=1}^{n}\sum_{j=1}^{n} z_i z_j \lim_{n\to\infty} k_n(x_i, x_j) \geq 0$$

$$\Rightarrow \sum_{i=1}^{n}\sum_{j=1}^{n} z_i z_j\, K(x_i, x_j) \geq 0$$

$$\Rightarrow z^T K z \geq 0$$

$\therefore K$ is +ve semi definite

$\therefore$ From Mercer's Theorem, K is a valid kernel.

$$k_n(x, y) = \langle \phi_n(x),\, \phi_n(y)\rangle = \phi_n(x)^T\, \phi_n(y)$$

Feature map for k_n, $\phi(x)$,

Define $\phi(x) = (\phi_1(x),\, \phi_2(x),\, \phi_3(x),\, \phi_4(x),\, \ldots)$

$$\langle \phi(x),\, \phi(y)\rangle = \sum_{n=1}^{\infty} \langle \phi_n(y),\, \phi_n(x)\rangle$$

$$\langle \phi(x),\, \phi(y) \rangle = \phi(x)^T \phi(y) = \sum_{n=1}^{\infty} \phi_n(x)^T \phi_n(y)$$

$$= \sum_{n=1}^{\infty} k_n(x,y) = k(x,y)$$

$\lim_{n\to\infty}$

$$\langle \phi(x),\, \phi(y) \rangle = \lim_{\substack{n\to\infty \\ \ell\to\infty}} \sum_i k_i(x,y) = k(x,y)$$

$\therefore$ Feature map for k,

$$\phi(x) = (\phi_1(x),\, \phi_2(x),\, \phi_3(x),\, \ldots)$$

(iv)

$$e^{k_n(x,y)} = \sum_{i=0}^{\infty} \frac{(k_n(x,y))^i}{i!}$$

Since product of 2 kernels is also a kernel, $(k_n(x,y))^i$ is a valid kernel. [$\because$ Part (ii)]
Since $m! > 0$, $\dfrac{1}{m!}(k_n(x,y))^i$ is a valid kernel. [$\because$ Part (i)]
Since sum of kernels is a valid kernel,

$$\sum_{i=0}^{\infty} \frac{(k_n(x,y))^i}{i!} \quad \text{is a valid kernel} \qquad [\because \text{Part (i)}]$$

$$\|x-y\|^2 = \|x\|^2 + \|y\|^2 - 2x^T y$$

$$\|x-y\|^2 = (x-y)^T (x-y)$$

$$= (x^T - y^T)(x-y) = x^T x - x^T y - y^T x + y^T y$$

$$= \|X\|^2 + \|Y\|^2 - 2x^T y$$

$$\therefore \exp\left(-\frac{\|x-y\|^2}{2\sigma^2}\right) = \exp\left(\frac{-\|x\|^2}{2\sigma^2} - \frac{\|y\|^2}{2\sigma^2} + \frac{2x^T y}{2\sigma^2}\right)$$

Using $f(z) = e^{-\|z\|^2/2\sigma^2}$,

$$\exp\left(\frac{-\|x-y\|^2}{2\sigma^2}\right) = f(x) \cdot \exp\left(\frac{x^T y}{\sigma^2}\right) \cdot f(y)$$

$x^T y$ is a valid linear kernel.
So $\exp\left(\dfrac{x^T y}{\sigma^2}\right)$ is a valid kernel $\left[\because \dfrac{1}{\sigma^2} > 0\right]$
$f(x)$ & $f(y)$ a scale the kernel.
$\therefore f(x)\, f(y) \exp\left(\dfrac{x^T y}{\sigma^2}\right)$ is a valid kernel.
$\therefore \exp\left(\dfrac{-\|x-y\|^2}{2\sigma^2}\right)$ is a valid kernel.

Solution 3

Let's say a person's age now = a
So,

$$\frac{P(y|x=a+1)}{1-P(y|x=a+1)} = \left(1+\frac{1.75}{100}\right)\left(\frac{P(y|x=a)}{1-P(y|x=a)}\right)$$

$P(y|x=q)$ is $P(y=1|x=q)$ = the probability of a person being unemployed for more than a year given that they were fired at the age q.

$$\frac{P(y|x=a+1)}{1-P(y|x=a+1)} = \frac{\frac{1}{1+e^{-(a+1)w+b}}}{\frac{e^{-(a+1)w+b}}{1+e^{-(a+1)w+b}}}$$

$$= e^{(a+1)w+b}$$

Similarly $\frac{P(y|x=a)}{1-P(y|x=a)} = e^{aw+b}$
So,

$$e^{aw+w+b} = 1.0175e^{aw+b}$$

Taking log to the base e on both sides,

$$aw+w+b = \ln 1.0175 + aw + b$$

$$\Rightarrow w = \ln 1.0175$$

Also,

$$\frac{P(y|x=34)}{1-P(y|x=34)} = 0.5$$

$$\Rightarrow P(y|x=34) = 0.5 - P(y|x=34)0.5$$

$$1.5P(y|x=34) = 0.5$$

$$P(y|x=34) = \frac{0.5}{1.5} = \frac{1}{3}$$

$$P(y|x=34) = \frac{1}{1+e^{-34w-b}} = \frac{1}{3}$$

$$2 = e^{-34w-b} \Rightarrow \ln 2 = -34w - b$$

$$-\ln 2 = 34\ln 1.0175 + b$$

$$b = -(\ln 2 + 34\ln 1.0175)$$

$\therefore w = \ln 1.0175$
$b = -(\ln 2 + 34\ln 1.0175)$

Solution 4

We see that we get a good split (approximately equal) if we split on $w \geq 65$,

$$\boxed{x_2 \geq 65}$$

Now out of the there are 5 training examples that will move on to the right subtree and 3 will move on to the left subtree.

Among For the right subtree, if we now split on height ≥ 95, we get only P examples in the left subtree giving us a leaf and F examples & 1 P example in the right subtree.

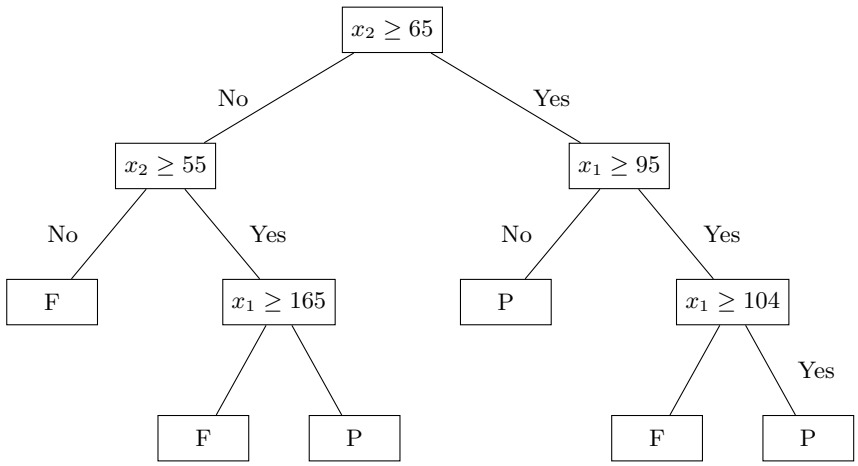
So in the right subtree we can take a split for height ≥ 104, giving us a clean split between P and F examples.

Now for left subtree where $x_2 < 65$ holds true, we have 1 P & 2 F examples:

$$(166, 56, P)\ ,\ (158, 62, F)\ ,\ (176, 52, F)$$

Taking the split $x_2 \geq 55$, isolates 1 F example in the left subtree further. Then we can take the split $x_1 \geq 165$ gives a clean split b/w the 2 examples.

Decision Tree:



This tree classifies all the 8 training examples correctly (so zero classification error).

It has 5 internal nodes & 6 leaf nodes.

So the least no. of nodes required to fit the data is $5 + 6 = 11$ nodes.

Solution 5

Probability that a given feature data point (of message) with features x is spam is:

$$P(y=1|x) = \frac{P(x|y=1)P(y=1)}{P(x|y=1)P(y=1) + P(x|y=0)P(y=0)} \quad [\because \text{Bayes Rule}]$$

y is the target label, $y \in \{0, 1\}$, where:

- $y = 1$ indicates that the message is spam.
- $y = 0$ indicates the message is not spam.

Our dictionary is of length 6. $x \in \mathbb{R}^6$ (the feature vector).

$$x_j = I\{\text{word } j \text{ appears in the message}\} \quad [\because I \text{ is the indicator function}]$$

Let the parameters be:

$$\begin{aligned} \phi_{j|y=1} &= P(x_j = 1|y=1) \quad [j \in \{1,2,3,4,5,6\}] \\ \phi_{j|y=0} &= P(x_j = 1|y=0) \\ \phi_y &= P(y=1) \end{aligned}$$

$\phi_{j|y=1}$ is the probability word j appears in the email given that the email is spam.

Calculations from Data

Total no. of emails $(m) = 200$.
No. of spam emails $= 50$.
No. of non-spam emails $= 150$.
So, $\phi_y = \frac{50}{200} = \frac{1}{4}$.

Parameters (Without Laplace Smoothing)

We know that using M.L.E (and from the Naive Bayes assumption that the x_i's are conditionally independent given y):

$$\phi_{j|y=1} = \frac{\sum_{i=1}^m I\{x_j^{(i)} = 1 \wedge y^{(i)} = 1\}}{\sum_{i=1}^m I\{y^{(i)} = 1\}}$$

$$\phi_{j|y=0} = \frac{\sum_{i=1}^m I\{x_j^{(i)} = 1 \wedge y^{(i)} = 0\}}{\sum_{i=1}^m I\{y^{(i)} = 0\}}$$

$$\phi_y = \frac{\sum_{i=1}^m I\{y^{(i)} = 1\}}{m}$$

For $y = 1$ (Spam): $\phi_{1|y=1} = \frac{20}{50} = \frac{2}{5}$, $\phi_{2|y=1} = \frac{40}{50} = \frac{4}{5}$, $\phi_{3|y=1} = \frac{30}{50} = \frac{3}{5}$, $\phi_{4|y=1} = 0$, $\phi_{5|y=1} = \frac{30}{50} = \frac{3}{5}$, $\phi_{6|y=1} = \frac{20}{50} = \frac{2}{5}$.

For $y = 0$ (Non-Spam): $\phi_{1|y=0} = \frac{15}{150} = \frac{1}{10}$, $\phi_{2|y=0} = \frac{15}{150} = \frac{1}{10}$, $\phi_{3|y=0} = \frac{15}{150} = \frac{1}{10}$, $\phi_{4|y=0} = \frac{60}{150} = \frac{2}{5}$, $\phi_{5|y=0} = \frac{45}{150} = \frac{3}{10}$, $\phi_{6|y=0} = \frac{30}{150} = \frac{1}{5}$.

Laplace Smoothing Parameters

With Laplace smoothing,

$$\phi_{j|y=1} = \frac{1 + \sum_{i=1}^m I\{x_j^{(i)} = 1 \wedge y^{(i)} = 1\}}{2 + \sum_{i=1}^m I\{y^{(i)} = 1\}} \tag{1}$$

$$\phi_{j|y=0} = \frac{1 + \sum_{i=1}^m I\{x_j^{(i)} = 1 \wedge y^{(i)} = 0\}}{2 + \sum_{i=1}^m I\{y^{(i)} = 0\}} \tag{2}$$

$$\phi_{1|y=1} = \frac{21}{52}, \quad \phi_{2|y=1} = \frac{41}{52}, \quad \phi_{3|y=1} = \frac{31}{52}$$

$$\phi_{4|y=1} = \frac{1}{52}, \quad \phi_{5|y=1} = \frac{31}{52}, \quad \phi_{6|y=1} = \frac{21}{52}$$

$$\phi_{1|y=0} = \frac{16}{152}, \quad \phi_{2|y=0} = \frac{16}{152}, \quad \phi_{3|y=0} = \frac{16}{152}$$

$$\phi_{4|y=0} = \frac{61}{152}, \quad \phi_{5|y=0} = \frac{46}{152}, \quad \phi_{6|y=0} = \frac{31}{152}$$

Sentence 1

From the Naive Bayes assumption,

$$P(x|y) = \prod_{j=1}^{6} P(x_j|y)$$

"This is free money meeting" So $x = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$

$$\begin{aligned} P(y=1|x) &= \frac{P(x_1, x_2, x_3, x_4, x_5, x_6|y=1)P(y=1)}{P(x|y=1)P(y=1) + P(x|y=0)P(y=0)} \\ &= \frac{(1-\phi_{1|y=1})\phi_{2|y=1}\phi_{3|y=1}\phi_{4|y=1}\phi_{5|y=1}\phi_{6|y=1}\phi_y}{P(x|y=1)P(y=1) + P(x|y=0)P(y=0)} \\ &= \frac{\frac{3\times4\times3\times0\times3\times2}{5^6} \times \frac{1}{4}}{0 + P(x|y=0)P(y=0)} \end{aligned}$$

$P(x|y=0)P(y=0) \neq 0$, so

$$P(y=1|x) = 0$$

With Laplace Smoothing

With Laplace smoothing,

$$\begin{aligned} P(y=1)P(x|y=1) &= \frac{31 \times 41 \times 31 \times 1 \times 31 \times 21}{52^6} \times \frac{1}{4} \\ &\approx 3.24 \times 10^{-4} \end{aligned}$$

$$\begin{aligned} P(y=0)P(x|y=0) &= \frac{136 \times 16 \times 16 \times 61 \times 46 \times 31}{152^6} \times \frac{3}{4} \\ &= 1.84 \times 10^{-4} \end{aligned}$$

So $P(y=1|x) = \frac{3.24\times10^{-4}}{5.09\times10^{-4}} = 6.365 \times 10^{-1}$

Sentence 2

$S_2 =$ Free this, win money!

$$X = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

Without Laplace smoothing,

$$P(y=1)P(x|y=1) = \frac{2\times4\times3\times1\times3\times3}{5^6}\times\frac{1}{4}$$
$$= 3.456\times10^{-3}$$

$$P(y=0)P(x|y=0) = \frac{1\times1\times1\times6\times3\times8}{10^6}\times\frac{3}{4}$$
$$= 108\times10^{-6} = 1.08\times10^{-4}$$

$$P(y=1|x) = \frac{3.456\times10^{-3}}{3.564\times10^{-3}} \approx 0.9697$$

With Laplace smoothing,

$$P(x|y=1)P(y=1) = \frac{21\times41\times31\times51\times31\times31}{52^6\times4}$$
$$= 0.0165$$

$$P(x|y=0)P(y=0) = \frac{16\times16\times16\times91\times46\times121\times3}{152^6\times4}$$

$$\approx 0.0001261$$

$$P(y=1|x) \approx \frac{0.0165}{0.01667} \approx 0.9924$$

Sentence 3

S_3 = Meeting is free.

$$X = \begin{bmatrix}0\\1\\0\\1\\0\\1\end{bmatrix}$$

Without Laplace Smoothing,

$$P(x|y=1)P(y=1) = 0 \quad [\because P(x_4=1|y=1)=0]$$

$P(x|y=0)P(y=0)$ is non-zero. So $P(y=1|x)=0$

With Laplace Smoothing,

$$P(x|y=1)P(y=1) = \frac{31\times41\times21\times1\times21\times21}{52^6}\times\frac{1}{4}$$
$$\approx 0.0001488$$

$$P(x|y=0)P(y=0) = \frac{136\times16\times136\times61\times106\times31\times3}{152^6\times4}$$
$$\approx 0.003607$$

$$P(y=1|x) = \frac{0.0001488}{0.003756}$$
$$\approx 0.039625$$